Set 11 at Daan — the front window where a cab would be.

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VAL256

Taipei's original 102-car medium-capacity fleet — selected in 1988, built in 1991, reshaped after two test fires and later converted from Matra fixed-block control to Bombardier Cityflo 650.

BR WENHU LINE [/RAIL/METRO/LINES/WENHU-LINE/] 文湖線

SYSTEM CONTRACTOR	CONTRACT	SELECTED	MANUFACTURED
Matra Transport [1]	CC350 [2]	March 1988 [1]	1991 [3]
ENTERED SERVICE	FLEET	SERVICE FORMATION	
28 March 1996 [3]	51 pairs / 102 cars [2]	Two pairs / four cars [3]	

Formation

2 CARS · 2 UNITS OF 1 · ▬ ONE UNIT ▬ COUPLING

Overview

DORTS selected the French Matra V256 system through an international competition in March 1988.[1] The builder's later technical history calls the fleet VAL 256, identifies it as a Matra-produced derivative of the VAL system and records manufacture in 1991.[3] The cars opened Taipei [/rail/thsr/stations/taipei/]'s first metro service on the Muzha Line [/rail/metro/lines/wenhu-line/] on 28 March 1996.[1] [3]

Contract CC350 covered 51 two-car pairs, or 102 cars.[2] DORTS describes Taipei's service pattern as two pairs coupled into a four-car train — the first such double-pair operation in this vehicle family.[3] The records recovered here do not say whether the same pairs remain permanently coupled into one four-car set, so train-level coupling permanence remains TBC.

Carbody and passenger layout

Each car is 13.78 m long, 2.56 m wide and 3.53 m high; DORTS says the width is the source of the `256` designation.[3] A car has two outside-sliding passenger doors on each side and no gangway to the next car.[3] [9] That separation means an evacuation between stations must use the side doors rather than an internal route through the train.[9]

TRTC [/rail/operators/trtc/]'s current network introduction describes a four-car Matra train as carrying approximately 420 people at five standing passengers per square metre.[13] That is not the fleet's original interior specification. In a 2023 trial on three trains, TRTC removed 12 seats and the luggage rack from the middle area of each of cars 2 and 3, added 16 hand straps and transverse handrails, narrowed door-side rails in cars 1 and 4, and removed selected doorway poles throughout the train.[12] The operator said those changes raised each trial train's approximate capacity from 400 to 420.[12]

Propulsion, automation and guideway

Each car has two separately excited, self-cooled DC traction motors with class-H insulation, and each motor armature drives through its own shaft.[8] The original system used automatic train operation without an onboard driver; DORTS published a minimum operating headway of 72 seconds.[5] TRTC now gives 80 km/h as the fleet's maximum speed and says normal operation is fully computer-controlled from the operations control centre, with manual driving available when required.[13]

The fleet runs on rubber tyres over dedicated running surfaces.[3] Most of the original Muzha [/rail/metro/stations/br02/] alignment used guided concrete surfaces, while the later network combines reinforced-concrete and steel running surfaces.[3] [4] [13] No opened primary vehicle specification in this search gave empty mass, maximum operating speed, motor rating or a complete braking specification, so those values are not reconstructed from the old encyclopedia table.

Commissioning fires

Two fires during testing burned the skins at car ends before opening.[7] DORTS's engineering account ties them directly to Taipei's novel four-car use: the original VAL arrangement contemplated one car or one two-car pair, but Matra had not coordinated propulsion and braking signals correctly across the two pairs.[7] One pair could continue driving while the other braked, leaving a brake calliper gripping its disc until heat ignited a rubber tyre.[7]

The same account says the underlying cause was not removed after the first fire, so a second occurred within six months.[7] DORTS then convened academic experts and Matra engineers, used simulation and verification to identify the cause, and modified the design.[7] This primary account replaces the former page's uncited shorthand and links the incident to the four-car formation itself.

Train-control conversion

The original VAL 256 train-control system used fixed blocks, with wayside control units and track transmission loops carrying train-position, speed, command and voice functions.[5] [6] The Neihu [/rail/metro/stations/br19/] extension was designed as a one-seat continuation of the Muzha railway, so the 51 old pairs and 101 new Bombardier pairs had to operate automatically over one line without a transfer.[6]

DORTS therefore replaced the VAL fleet's onboard control and communication equipment and installed Bombardier Cityflo 650 moving-block control.[6] The existing Muzha points and platform doors remained but were interfaced to the new system; after cutover, DORTS says the original Matra control equipment was completely removed.[6] TRTC records all 51 converted pairs joining Wenhu Line passenger service on 26 December 2010.[11]

Refurbishment

An operator-authored DORTS article records renewal of the VAL 256 air-conditioning plant, replacement of T12 saloon lamps and tail lights with LEDs, and addition of automatic temperature control that could raise the setting by 1°C off peak.[10] The opened source describes these as energy-saving measures but does not provide unit-by-unit completion dates or contract values. [10]

The signed procurement documents, acceptance chronology, current vehicle asset register and any awarded replacement contract were not recovered in this search. Builder allocation beyond Matra, exact delivery dates by pair, present depot allocation and a confirmed withdrawal programme therefore remain TBC.

Specifications

Car length	13.78	m [3]
Car width	2.56	m [3]
Car height	3.53	m [3]
Maximum speed	80	km/h [13]
Passenger doors	4	per car, 2 per side [3]
Traction motors	2	separately excited DC motors per car [8]
Train capacity	approximately 420	at 5 standing passengers/m² after interior changes [13]
Train control	Cityflo 650 moving block	[6]
Inter-car gangway	None	[9]
Empty mass	TBC	t
Maximum operating speed	TBC	km/h
Current depot allocation	TBC	

References

13 of 13 sources on this page are primary — published by the operator, the builder or government.

1 Mr Pao-Cheng Chi and Taipei MRT artefacts [https://data.taipei/api/dataset/3812b03c-7e72-40c7-85cf-c67df9162f7e/resource/75f441a8-9cbc-4bf6-8b8b-e8cee765b369/download] 齊寶錚先生與臺北捷運文物
PRIMARY Taipei City Government Department of Rapid Transit Systems (臺北市政府捷運工程局) accessed 2026-08-23
DORTS dates the international competition and selection of Matra's V256 system to March 1988, and dates opening to 28 March 1996.

2 DORTS frequently asked questions — train quantities [https://www.dorts.gov.taipei/News_Content.aspx?n=2A66A485FACB0D5B&s=C8602F8588914E91]
機電設計－台北都會區大眾捷運系統路網中，各路線所採購之列車數為何？
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS gives contract CC350 and 51 pairs / 102 cars for the original Muzha Line.

3 The development of Taipei Metro electromechanical planning and design — VAL 256 [https://ebook.dorts.gov.taipei/JRTST/ebook/no48/files/basic-html/page138.html]
臺北捷運機電系統精進軌跡尋蹤－機電規設之蛻變、創新與成長
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS identifies Matra, 1991 manufacture, the four-car service arrangement, opening date, principal dimensions, door count, automation and concrete running surface.

4 The development of Taipei Metro electromechanical planning and design — Wenhu conversion [https://ebook.dorts.gov.taipei/JRTST/ebook/no48/files/basic-html/page139.html]
臺北捷運機電系統精進軌跡尋蹤－機電規設之蛻變、創新與成長
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS records conversion of the original VAL 256 train-control equipment to Bombardier's system after the Neihu Line opened and describes the concrete/steel running surfaces.

5 Review and prospects for Taipei Metro electromechanical systems — original Muzha system [https://ebook.dorts.gov.taipei/JRTST/ebook/no48/files/basic-html/page202.html]
臺北捷運機電工程回顧與展望
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS describes the automatic rubber-tyred Matra system, two-pair operation, 72-second minimum headway and original fixed-block control.

6 Review and prospects for Taipei Metro electromechanical systems — Wenhu integration [https://ebook.dorts.gov.taipei/JRTST/ebook/no48/files/basic-html/page203.html]
臺北捷運機電工程回顧與展望
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS identifies the 51 old pairs, onboard-equipment replacement, Cityflo 650 moving block, retained points and platform doors, and removal of the old Matra train-control equipment.

7 A proposal for DORTS knowledge management and knowledge upgrading [https://ebook.dorts.gov.taipei/JRTST/ebook/no50/files/basic-html/page56.html]
捷運局知識管理與知識升級芻議
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23

DORTS records two test fires, explains the two-pair propulsion/braking synchronisation defect it found and says the design was corrected after expert simulation and verification.

8 Electrical equipment of electric multiple units [https://ebook.dorts.gov.taipei/JRTST/ebook/no35/files/basic-html/page34.html]
電聯車電氣設備
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS specifies two separately excited, self-cooled, class-H insulated DC traction motors per Matra car.

9 Practical guide to metro electric multiple units — inter-car gangways [https://ebook.dorts.gov.taipei/ebook/no21/files/basic-html/page112.html]
捷運電聯車實務
PRIMARY Taipei City Government Department of Rapid Transit Systems（ 臺北市政府捷運工程局 ）
accessed 2026-08-23
DORTS states that VAL 256 cars are fully separated without inter-car gangways, so evacuation uses side doors.

10 Taipei Metro Corporation's energy-saving and carbon-reduction measures [https://ebook.dorts.gov.taipei/JRTST/ebook/no50/files/basic-html/page48.html]
臺北捷運公司節能減碳作法
PRIMARY
Taipei City Government Department of Rapid Transit Systems / Taipei Rapid Transit Corporation（ 臺北市政府捷運工程局／臺北大眾捷運股份有限公司 ） accessed 2026-08-23
The operator-authored article records VAL 256 air-conditioning renewal, T12-to-LED saloon lighting, LED tail lights and automatic temperature control.

11 Taipei Metro 2010 annual report — significant events [https://data.taipei/api/dataset/6ceca6ae-05ec-4995-bc1c-d3cd15f79ee1/resource/b04ee75d-55d8-4291-a96a-418f62c4e8a5/download]
臺北捷運公司99年年報－重要紀事
PRIMARY Taipei Rapid Transit Corporation（ 臺北大眾捷運股份有限公司 ） accessed 2026-08-23
TRTC records that all 51 converted VAL 256 pairs joined Wenhu Line operation on 26 December 2010.

12 Taipei Metro trials interior optimisation on three Matra trains [https://www.metro.taipei/News_Content.aspx?n=30CCEFD2A45592BF&sms=72544237BBE4C5F6&s=86F98235BC105C44]
臺北捷運化身空間魔術師！繼龐巴迪列車改善後 文湖線馬特拉列車空間優化
PRIMARY Taipei Rapid Transit Corporation（ 臺北大眾捷運股份有限公司 ） accessed 2026-08-23
TRTC describes the 2023 three-train trial, exact seat and luggage-rack removals, added handholds and the capacity change from approximately 400 to 420.

13 Taipei Metro network introduction — Wenhu Line vehicles [https://www.metro.taipei/cp.aspx?n=ccf30033e6ed8008&s=405306CC7ACD137B]
路網簡介－文湖線介紹
PRIMARY Taipei Rapid Transit Corporation（ 臺北大眾捷運股份有限公司 ） accessed 2026-08-23

The operator publishes the two-pair/four-car arrangement, 80 km/h maximum and approximately 420-passenger Matra capacity at five standing passengers per square metre.

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Station names, codes and sequence from TDX Transport Data eXchange, Ministry of Transportation and Communications, operators TRTC, NTMC, TYMC, and TMRT. Government open data. Retrieved 2026-08-24; source last updated 2016-11-23. The 12 Sanying station records absent from TDX are sourced directly to New Taipei Metro and government publications on their pages. Provenance.